

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks: 50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I S. Santhiya (19251136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. 

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.

- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
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Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

	Problem Identification (20%)	Debugging (20%)	Optimization (25%)	Analysis (20%)	Code Quality (15%)
Q1	1.5	2	1.5	2	1 8
Q2	1	2	1	2	1 7
Q3	1.5	2.5	1.5	1.5	1 8
Q4	2	1.5	2	1.5	1.5 8.5
Q5	2	2	2	2	1.5 9.5

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)

		understandin g(2)			
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

SIMATS ENGINEERING
ASSESSMENT TOOL 3

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COURSE CODE: CSA0720

COURSE OUTCOME COVERED

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Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5 **Total Marks:50**

Annexure A

Debugging & Optimisation Task

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Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- **Identifying the subnet to which the configured IP address belongs.**
- **Determining the network address, broadcast address, and valid host range.**
- **Verifying whether the configured IP address and gateway are valid.**
- **Identifying the configuration errors.**
- **Optimizing the configuration by assigning an appropriate IP address and default gateway.**
- **Calculating the total number of usable host addresses after correction.**

Answers:

Given

- **IP Address = 192.168.10.65**
- **Subnet Mask = 255.255.255.192 (/26)**
- **Default Gateway = 192.168.10.1**

Step 1: Find the block size

Last octet of subnet mask = 192

Block size:

$$256 - 192 = 64$$

So, the subnets are:

- **192.168.10.0 – 192.168.10.63**
- **192.168.10.64 – 192.168.10.127**
- **192.168.10.128 – 192.168.10.191**
- **192.168.10.192 – 192.168.10.255**

Step 2: Identify the subnet

The IP address is 192.168.10.65.

It falls between 192.168.10.64 and 192.168.10.127.

Therefore:

Subnet = 192.168.10.64/26

Step 3: Find the network address

The first address of the subnet is:

Network Address = 192.168.10.64

Step 4: Find the broadcast address

The last address of the subnet is:

Broadcast Address = 192.168.10.127

Step 5: Find the valid host range

Network address and broadcast address cannot be assigned to hosts.

Therefore:

First Host = 192.168.10.65

Last Host = 192.168.10.126

So:

Valid Host Range = 192.168.10.65 – 192.168.10.126

Step 6: Check the IP address

Configured IP:

192.168.10.65

It is within the valid host range.

Therefore:

IP Address = Valid

Step 7: Check the default gateway

Configured gateway:

192.168.10.1

The gateway 192.168.10.1 belongs to the subnet:

192.168.10.0/26

But the workstation belongs to:

192.168.10.64/26

Step 8: Correct the configuration

The gateway must belong to the same subnet.

Correct configuration:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.10.66

Step 9: Calculate usable host addresses

For /26:

Host bits = $32 - 26 = 6$

Total addresses:

$2^6 = 64$

Usable addresses:

$64 - 2 = 62$

2. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Answers:

Given

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand Device A

IPv6 address:

2001:db8::ff00:42:8329

Expanded form:

2001:0db8:0000:0000:ff00:0042:8329

There are only 7 groups written above, so the correct complete expansion is:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 2: Expand Device B

IPv6 address:

2001:db8:0:0:1::1

Expanded form:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 3: Find the network prefix

Prefix length = /64

Therefore, the first 64 bits (first 4 groups) represent the network.

Device A:

2001:0db8:0000:0000::/64

Device B:

2001:0db8:0000:0000::/64

Step 4: Check whether they are in the same subnet

Device A prefix:

2001:db8:0:0::/64

Device B prefix:

2001:db8:0:0::/64

Both prefixes are the same.

Therefore:

Both devices belong to the same subnet.

Step 5: Identify the configuration mismatch

There is no /64 prefix mismatch.

The addresses are:

- Device A: 2001:db8:0:0:0:ff00:42:8329/64
- Device B: 2001:db8:0:0:1:0:0:1/64

The difference occurs after the first 64 bits, so it does not place them in different /64 subnets.

Therefore, the communication failure cannot be caused by a subnet-prefix mismatch based on these addresses alone.

Step 6: Appropriate IPv6 configuration

Both addresses can remain in the same subnet:

- **Device A:** 2001:db8:0:0::ff00:42:8329/64
- **Device B:** 2001:db8:0:0::1/64

Network prefix:

2001:db8:0:0::/64

Step 7: Calculate available host addresses

For IPv6 /64:

Host bits:

$$128 - 64 = 64 \text{ bits}$$

Number of addresses:

$$2^{64}$$

Therefore:

$$\text{Available addresses} = 18,446,744,073,709,551,616$$

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- **Identifying the network address, broadcast address, and host range for each subnet.**
- **Calculating the number of usable and unused IP addresses.**
- **Identifying inefficient subnet allocations.**
- **Debugging the addressing plan by locating subnet allocation mistakes.**
- **Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.**

Answers:

Given

Network = 192.168.1.0/24

Subnet masks used:

- /26
- /27
- /28

Step 1: /26 Subnet

For /26:

Block size = $256 - 192 = 64$

Subnet 1

- Network Address: 192.168.1.0
- Broadcast Address: 192.168.1.63
- Host Range: 192.168.1.1 – 192.168.1.62
- Usable Hosts: 62

Subnet 2

- Network Address: 192.168.1.64
- Broadcast Address: 192.168.1.127
- Host Range: 192.168.1.65 – 192.168.1.126
- Usable Hosts: 62

Step 2: /27 Subnet

For /27:

Block size = $256 - 224 = 32$

Subnet 3

- Network Address: 192.168.1.128
- Broadcast Address: 192.168.1.159
- Host Range: 192.168.1.129 – 192.168.1.158
- Usable Hosts: 30

Subnet 4

- Network Address: 192.168.1.160
- Broadcast Address: 192.168.1.191
- Host Range: 192.168.1.161 – 192.168.1.190
- Usable Hosts: 30

Step 3: /28 Subnet

For /28:

Block size = $256 - 240 = 16$

Subnet 5

- Network Address: 192.168.1.192
- Broadcast Address: 192.168.1.207
- Host Range: 192.168.1.193 – 192.168.1.206
- Usable Hosts: 14

Subnet 6

- Network Address: 192.168.1.208
- Broadcast Address: 192.168.1.223
- Host Range: 192.168.1.209 – 192.168.1.222
- Usable Hosts: 14

Step 4: Remaining addresses

The remaining range is:

192.168.1.224 – 192.168.1.255

This provides another /27:

- Network: 192.168.1.224
- Broadcast: 192.168.1.255
- Host Range: 192.168.1.225 – 192.168.1.254
- Usable Hosts: 30

Step 5: Identify inefficient allocation

- /26 provides 62 usable addresses.
- /27 provides 30 usable addresses.
- /28 provides 14 usable addresses.

If a small department needs only 10–12 addresses, assigning /26 is inefficient because many addresses remain unused.

If a department needs more than 30 addresses, /27 is insufficient.

Therefore, subnet size should be selected according to the actual number of hosts.

Step 6: Debugging mistakes

The main allocation mistakes are:

1. Giving /26 to small departments causes address wastage.
2. Giving /28 to departments requiring more than 14 hosts causes insufficient addresses.
3. Fixed-size subnetting does not efficiently match different department requirements.

Step 7: Optimized VLSM scheme

Using VLSM, assign larger subnets to departments with more hosts and smaller subnets to departments with fewer hosts.

Example:

Department	Required Host	VLSM Subnet	Usable host
Department A	50	192.168.1.0/26	62
Department B	25	192.168.1.64/27	30
Department C	12	192.168.1.96/28	14
Department D	6	192.168.1.112/29	6

4. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- **Expanding both IPv6 addresses into their complete notation.**
- **Determining the network prefix of each device.**
- **Verifying whether both devices belong to the same subnet.**
- **Identifying the addressing or prefix mismatch.**
- **Optimizing the IPv6 configuration by assigning appropriate addresses.**
- **Calculating the number of available host addresses within the corrected subnet.**

Answers:

Given

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand Device A

Given:

2001:db8::ff00:42:8329

Complete notation:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 2: Expand Device B

Given:

2001:db8:0:0:1::1

Complete notation:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 3: Find the network prefix

Since the prefix is /64, the first 4 groups represent the network.

Device A:

2001:0db8:0000:0000::/64

Device B:

2001:0db8:0000:0000::/64

Step 4: Check whether they belong to the same subnet

Both devices have the same first 64 bits:

2001:0db8:0000:0000

Therefore:

Both devices belong to the same subnet.

Step 5: Identify the mismatch

There is no IPv6 prefix mismatch.

- Device A = 2001:db8:0:0::/64
- Device B = 2001:db8:0:0::/64

The difference is in the host portion, which is allowed.

Therefore, the given addresses should be able to communicate at the subnet level.

Step 6: Optimized configuration

Use the same /64 network for both devices.

- **Device A:** 2001:db8:0:0::ff00:42:8329/64
- **Device B:** 2001:db8:0:0::1/64

Network prefix:

2001:db8:0:0::/64

Step 7: Calculate available host addresses

IPv6 has 128 bits.

Prefix = 64 bits

Host bits:

$128 - 64 = 64$ bits

Number of addresses:

$2^{64} = 18,446,744,073,709,551,616$

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answers:

Step 1: Check the destination IP

Destination:

192.168.2.100

It belongs to the network:

192.168.2.0/24

Because /24 means the first three octets identify the network:

192.168.2.xxx

Step 2: Find the matching route

Routing table contains:

192.168.2.0/24 → 10.0.0.3

Therefore, this is the matching entry.

Step 3: Check the Second Routing Entry

The second routing entry is:

192.168.2.0/24 → 10.0.0.3

The destination address is:

192.168.2.100

The first three octets are:

192.168.2

Therefore, the destination belongs to:

192.168.2.0/24

So the second routing entry is a valid match.

Step 4: Verify the Destination Subnet

For the subnet:

192.168.2.0/24

The subnet mask is:

255.255.255.0

The addresses are:

- Network Address = 192.168.2.0
- First Host = 192.168.2.1
- Last Host = 192.168.2.254
- Broadcast Address = 192.168.2.255

The destination address is:

192.168.2.100

Since it lies between 192.168.2.1 and 192.168.2.254, it is a valid host address.

Therefore:

192.168.2.100 belongs to the intended subnet.

Step 5: Check the Default Route

The routing table also contains:

Default Route → 10.0.0.1

The default route can be represented as:

0.0.0.0/0

It matches all IPv4 destinations, but it has the least specific prefix.

The router will use the default route only when there is no more specific route available.

Since 192.168.2.0/24 already matches the destination, the default route is not selected.

Step 6: Routing error

The routing entry is correct.

If communication fails, check:

- 10.0.0.3 connectivity
- Next router's routing table
- Interface configuration
- Return route

Step 7: Default route

If no specific route matches, the router uses:

0.0.0.0/0 → 10.0.0.1

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